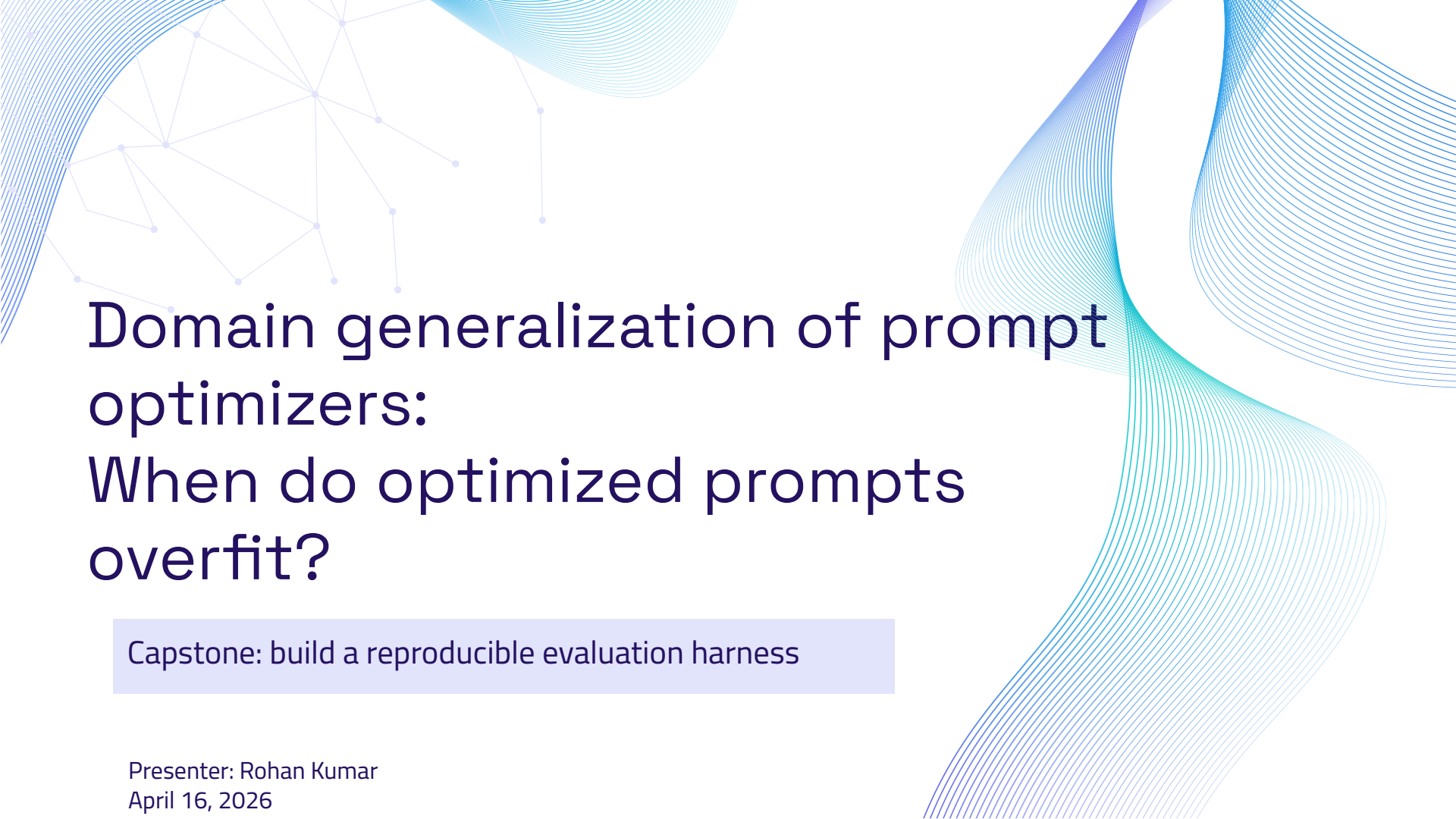




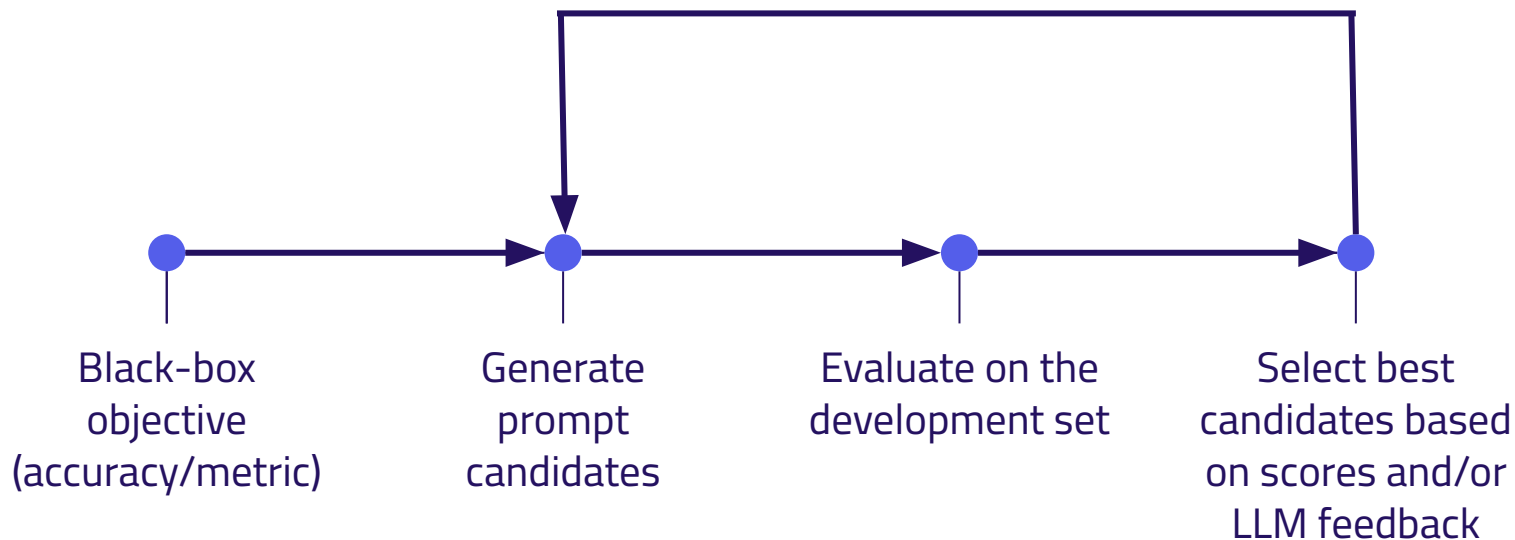
Domain generalization of prompt optimizers: When do optimized prompts overfit?

Capstone: build a reproducible evaluation harness

Why now: prompt optimization is a real subfield

-  APO techniques are proliferating
-  Strong gains but unclear transfer
-  Need systematic evaluation.

What is a prompt optimizer, concretely?



Representative optimizers we can compare

- EvoPrompt: evolutionary search
- TextGrad: textual gradients
- GEPA: reflective evolution with few rollouts
- ORPO: Optimization by PROMpting

Prior work on APO generalization

- Prompt domain generalization is understudied
- Limited prior work (NeurIPS 2024)
- Need a systematic re-evaluation using recent models and API methods

Potential research questions

- Q1: How large is the OOD gap across optimizers?
- Q2: Which prompts look robust (variance / worst-case)?
- Q3: More generally: which factors affect generalization?
- Q4: Stretch: can we improve OOD via simple procedures?

Experimental protocol

- Pick a source domain dataset with a dev split
- Optimize an instruction/prompt for high dev accuracy under a fixed budget of optimization steps
- Evaluate the same optimized prompt on (i) the source test split and (ii) one or more OOD target domains for the same nominal task.
- Stretch: A cross-model setting where we optimize for one model and test prompts on another

Potential datasets / tasks

- Classification (Fast iteration baseline)
 - NLI (e.g., MultiNLI)
 - Sentiment (e.g., Amazon / SST)
- Reasoning / Structured Tasks
 - Math word problems (e.g., GSM8K-style)
 - Logical reasoning / QA
- Code / Formal Tasks (Stretch but high value)
 - MBPP/HumanEval
 - NL2SQL (e.g., Spider 1.0, BIRD)

Deliverables

- Unified API for prompt optimizers
- Reproducible logs + costs
- Concrete recommendations and report

Thanks!

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